FINITELY MANY MAXIMAL IDEALS WITH LOCALLY NILPOTENT JACOBSON RADICAL MAKE A RING A PRODUCT OF ITS LOCALISATIONS

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Allowed standard prerequisites (editorial).

- Primes contain nilpotent elements. Used in the counted proof at lines 12987-12989. Shows every prime contains the intersection of the maximal ideals.
- Disjoint supports give a product decomposition. Invoked in the counted proof at line 12996. Splits the ring into a product of rings with singleton spectra, applied repeatedly.

Source reference. Original lemma 00EM: sources/stacks-algebra/pinned-original-source.tex, source lines 4602–4602. Navigation pointer for \ref{lemma-disjoint-implies-product} used inside the counted span (line 12996); the argument that consumes it is retained in the span.

Lemma 0.1. *Any ring with finitely many maximal ideals and locally nilpotent Jacobson radical is the product of its localizations at its maximal ideals. Also, all primes are maximal.*

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Proof. Let R be a ring with finitely many maximal ideals $\mathfrak{m}_1, \ldots, \mathfrak{m}_n$. Let $I = \bigcap_{i=1}^n \mathfrak{m}_i$ be the Jacobson radical of R. Assume I is locally nilpotent. Let $\mathfrak{p}$ be a prime ideal of R. Since every prime contains every nilpotent element of R we see $\mathfrak{p} \supset \mathfrak{m}_1 \cap \ldots \cap \mathfrak{m}_n$. Since $\mathfrak{m}_1 \cap \ldots \cap \mathfrak{m}_n \supset \mathfrak{m}_1 \ldots \mathfrak{m}_n$ we conclude $\mathfrak{p} \supset \mathfrak{m}_1 \ldots \mathfrak{m}_n$. Hence $\mathfrak{p} \supset \mathfrak{m}_i$ for some i, and so $\mathfrak{p} = \mathfrak{m}_i$. Thus the spectrum of R is the discrete topological space $\{\mathfrak{m}_1, \ldots, \mathfrak{m}_n\}$. By Lemma 00EM applied $n-1$ times we find that $R = R_1 \times \ldots \times R_n$ where the spectrum of R_i is a singleton for each i. Thus R_i is a local ring and since it is a localization of R (by the lemma), it is one of the local rings of R as desired. $\square$